

Thermodynamic approach and Saddle-Dominated Capacity of the LSE Dense Associative Memory

Anonymous submission

Abstract

The Dense Associative Memory with log-sum-exp (LSE) kernel on the N -sphere stores $M = \exp(\alpha N)$ patterns at load α . A Gaussian approximation of the inter-pattern alignments gives a zero-temperature capacity $\alpha_c = 1/2$, independent of the kernel sharpness β , the Gaussian-ensemble threshold of Lucibello and Mézard (2024). Using the exact spherical alignment density, we build a finite-temperature theory of retrieval. The sum over non-target patterns is then set by an interior saddle rather than by the right tail of the Gaussian, which restores the β -dependence: at $T = 0$ the capacity grows from $\sim \beta$ at small β to $\frac{1}{2} \ln \beta$ at large β , is ≈ 0.6226 at $\beta = 1$, and reproduces the exact spherical threshold of Lucibello and Mézard. At $T > 0$ the retrieval basin is bounded by a single-pattern spinodal, beyond which it no longer exists; below it the basin is metastable, with a Kramers escape time exponential in N . We confirm the resulting (α, T) stability diagram by Monte Carlo: the Full sampler stores all M patterns but reaches only $N \sim 25$, the Cone sampler that keeps only the patterns in a cone around the target reaches $N \sim 30$, and a pattern-free Bulk sampler that follows the leading-order free-energy profile reaches $N \sim 10^4$. All three agree with the predicted spinodal to within the residual finite- N gap.

1 Introduction

Modern dense associative memory (DAM) models begin with Krotov and Hopfield (2016), who generalized the classical Hopfield network (Hopfield 1982) from pairwise to polynomial interactions, raising the storage capacity from linear to polynomial in the number of neurons N while preserving the ability to retrieve previously stored information from noisy or corrupted input data. Taking the polynomial degree to infinity yields an exponential capacity (Demircigil et al. 2017): for binary neurons (state space $\{-1, +1\}^N$), the limiting capacity is $M_c = \exp(\alpha_c N)$ with load $\alpha_c = (\ln 2)/2 \approx 0.35$.

The same exponential scaling carries over to continuous states. Ramsauer et al. (2021) introduced a modern Hopfield network whose state is a continuous vector in $\mathbb{R}^N$ and whose energy is a log-sum-exp (LSE) of the state-pattern alignments. We take both the stored patterns ξ^μ and the retrieval state x on the sphere $|x|^2 = N$, and store $M = \exp(\alpha N)$

patterns at load α . The energy then reads

$$H(x) = -\beta^{-1} \ln \sum_{\mu=1}^M \exp[-\beta N(1 - \phi^\mu)], \quad (1)$$

where β is a kernel sharpness controlling how strongly the energy distinguishes patterns, and ϕ^μ is the *alignment* of the current state x with the μ -th pattern ξ^μ ,

$$\phi^\mu(x) = N^{-1} x \cdot \xi^\mu. \quad (2)$$

All these models are energy-based. What sets the LSE energy apart from the classical, polynomial, and exponential models is the logarithm: it acts as a smooth maximum, keeping the energy extensive in N rather than exponentially large, while its gradient is a softmax over patterns, so the retrieval update replaces the state by a softmax-weighted average of the stored patterns, exactly like the attention mechanism of transformers. Model (1) is the Gaussian-kernel member of a family of kernel-based energies. Replacing the Gaussian by the compactly supported Epanechnikov kernel gives the log-sum-ReLU (LSR) model (Hoover et al. 2025).

A natural question is how retrieval survives thermal noise. At a finite temperature T of the thermal bath (distinct from the kernel sharpness β), the boundary $\alpha_c(\beta, T)$ in the load-temperature plane separates cues that relax back to their target from cues that escape to spurious states. Petrova, Polyachenko, and State (2026) derived this boundary for the LSE and LSR kernels under a Gaussian approximation of the inter-pattern alignment density, obtaining for LSE a β -independent zero-temperature load $\alpha_c = 1/2$, and identifying the Gaussian approximation as the likely source of error. Brute-force Monte Carlo (Petrova 2026) then showed deviations from the Gaussian prediction, with the retrieval region extending to larger α as $T \rightarrow 0$.

Here we revisit the LSE capacity on the N -sphere using the exact spherical alignment density, both analytically and by Monte Carlo, across the full (α, T) plane. The sum over non-target patterns is governed by an interior saddle of the partition-sum integrand rather than by the right edge of the alignment distribution, which restores the β -dependence missing from the Gaussian estimate. At $T = 0$ the resulting capacity reproduces the exact spherical threshold of Lucibello and Mézard (2024) and lies above the rigorous storage lower bound of Ramsauer et al. (2021). At $T > 0$ the

retrieval well is bounded by a spinodal, the limit of metastability beyond which the well disappears; below it the well persists metastably, and a trapped cue escapes only over a barrier, with a lifetime exponential in N , a Kramers escape (Kramers 1940; Hänggi, Talkner, and Borkovec 1990). This resolves the low- T deviations seen in simulations.

The paper is organized as follows. Section 2 builds the finite-temperature retrieval theory from the exact spherical alignment density: the Gaussian baseline and why it fails, the interior saddle that fixes the noise floor, the zero-temperature capacity, and the finite-temperature spinodal. The Monte Carlo schemes section 3 describes the Full, Cone, and Bulk samplers. The Results section presents the stability diagram in the (α, T) plane at $\beta = 1$, compares it against the Gaussian and spinodal predictions, and examines the finite- N behavior. The Discussion section 5 collects the implications and the relation to the literature.

2 Retrieval at large N

Let the retrieval pattern ξ^1 and a spurious pattern ξ be drawn independently and uniformly from the sphere of radius $\sqrt{N}$ in $\mathbb{R}^N$. The alignment

$$\varphi(\xi) \equiv N^{-1} \xi^1 \cdot \xi \quad (3)$$

is the cosine of the angle between them. By rotational symmetry we may fix the first pattern along the coordinate axis e_1 and treat the spurious pattern as a unit vector u uniformly distributed on S^{N-1} . The alignment is then simply its first coordinate, $\varphi = u_1$, and we seek the density of u_1 induced by the uniform measure on the sphere. The probability that u_1 lies in $[\varphi, \varphi + d\varphi]$ equals the area of the corresponding spherical band divided by the total area of S^{N-1} . Decomposing $u = (u_1, u_\perp)$, the constraint $|u| = 1$ fixes $|u_\perp| = \sqrt{1 - \varphi^2}$, so the points at height φ form a sphere S^{N-2} of radius $\sqrt{1 - \varphi^2}$, with surface area $\propto (1 - \varphi^2)^{(N-2)/2}$. The band thickness is $d\varphi/\sqrt{1 - \varphi^2}$, since the surface is inclined with respect to the axis. Normalizing the band area $\propto (1 - \varphi^2)^{(N-3)/2} d\varphi$ gives the density

$$h(\varphi) = C_N (1 - \varphi^2)^{(N-3)/2} \quad (4)$$

with the normalization constant

$$C_N = \frac{\Gamma(N/2)}{\sqrt{\pi} \Gamma((N-1)/2)}. \quad (5)$$

The density is symmetric about $\varphi = 0$, with mean zero and variance $1/N$. As $N \rightarrow \infty$ it concentrates near the equator and approaches a Gaussian,

$$h(\varphi) \longrightarrow \sqrt{\frac{N}{2\pi}} e^{-N\varphi^2/2}, \quad (6)$$

so distinct high-dimensional patterns are nearly orthogonal: typical alignments are of order $N^{-1/2}$. The exact law is retained where the rare large-alignment tail matters; the Gaussian form is adequate only near the center. The density (4) and its Gaussian limit are the spherical and Gaussian rate functions of Lucibello and Mézard (2024).

Extreme-value attempts. Ramsauer et al. (2021) provide two related lower bounds on the storage capacity, both based on extreme-value statistics of the inter-pattern alignments. Their Theorem 4 ties retrieval to the separation of a pattern from its nearest neighbour, $N(1 - \varphi_{\max})$, with $\varphi_{\max}$ the highest alignment among the $M = e^{\alpha N}$ patterns: retrieval fails when a spurious pattern reaches the target, $\varphi_{\max} \rightarrow 1$. The load at which this happens is fixed by the alignment law $h(\varphi)$ near $\varphi = 1$. Under the Gaussian approximation h keeps weight at $\varphi = 1$, and $\varphi_{\max}$ reaches 1 at the β -independent load $\alpha_c^G = 1/2$. Under the exact law (4) h vanishes at $\varphi = 1$, since two distinct patterns cannot align perfectly on the sphere; then

$$\varphi_{\max} = \sqrt{1 - e^{-2\alpha}} < 1 \quad \text{for every } \alpha, \quad (7)$$

and the capacity is infinite.

Their Theorem 3 acts directly on the softmax sum $S(x)$ from (1), which splits into a retrieval term and a sum over spurious patterns,

$$S(x) \equiv \exp[\beta N(\varphi(x) - 1)] + S_{\text{spur}}(x), \quad (8)$$

$$S_{\text{spur}}(x) \equiv \sum_{\mu=2}^M \exp[\beta N(\varphi^\mu(x) - 1)]. \quad (9)$$

The bound replaces S_{spur} by M times its largest term,

$$S_{\text{spur}} \leq M e^{-\beta N(1 - \varphi_{\max})} = e^{N(\alpha - \beta(1 - \varphi_{\max}))}, \quad (10)$$

which exceeds the retrieval term once $\varphi_{\max} > 1 - \alpha/\beta$. With $\varphi_{\max}$ from (7), Ramsauer et al.'s capacity solves

$$\sqrt{1 - e^{-2\alpha}} = 1 - \frac{\alpha}{\beta}, \quad (11)$$

finite and β -dependent.

Neither bound is tight. Theorem 4 with the Gaussian gives a β -independent $1/2$; with the exact density it gives infinity. Theorem 3 gives a finite β -dependent capacity but assumes S_{spur} is dominated by its largest term. The saddle-point analysis below evaluates S_{spur} without that approximation.

Saddle-point evaluation. The spurious patterns are i.i.d. uniform on the sphere and independent of x , so for any x with $|x|^2 = N$ the projections $\varphi^\mu(x) = x \cdot \xi^\mu/N$ have marginal density (4). The spurious sum therefore has the same distribution at every x on the sphere; in particular at $x = \xi^1$, where the projections become the inter-pattern alignments $\varphi_{1\mu}$. With $M = e^{\alpha N}$, the sum concentrates around its expectation, and

$$S_{\text{spur}} \approx M \int_{-1}^1 d\varphi h(\varphi) \exp[\beta N(\varphi - 1)]. \quad (12)$$

The large- N limit of (12) is set by a saddle point. The saddle φ^* solves $\partial g(\varphi, \beta)/\partial \varphi = 0$, with

$$g(\varphi, \beta) = \ln(1 - \varphi^2)/2 + \beta \varphi. \quad (13)$$

This gives the saddle point at

$$\varphi^*(\beta) = \frac{\sqrt{1 + 4\beta^2} - 1}{2\beta}, \quad (14)$$

and the saddle value

$$g_{\max}(\beta) = g(\varphi^*, \beta) = \frac{1}{2} \ln(1 - \varphi^{*2}) + \beta \varphi^*. \quad (15)$$

The leading-order spurious sum is then

$$S_{\text{spur}} \sim \exp[N(\alpha + g_{\max}(\beta) - \beta)]. \quad (16)$$

Appendix A derives the prefactor; it depends only on $\varphi^*(\beta)$ and equals 1.082 at $\beta = 1$. In the large- N limit only the leading term in the exponent (proportional to N) survives, so the prefactor is not needed at this stage.

Zero-temperature capacity. The internal energy density $u = H/N$ from (1) reads

$$u(\varphi) = \begin{cases} 1 - \varphi & \text{retrieval term dominates,} \\ 1 - [\alpha + g_{\max}(\beta)]/\beta & \text{spurious term dominates.} \end{cases} \quad (17)$$

The two branches meet in a cusp at

$$\varphi_c(\alpha, \beta) = (\alpha + g_{\max}(\beta))/\beta, \quad (18)$$

where dominance switches. At zero temperature the retrieval minimum sits at $\varphi = 1$; it is the global minimum as long as $\varphi_c < 1$, giving the zero-temperature capacity

$$\alpha_c(\beta, 0) = \beta - g_{\max}(\beta). \quad (19)$$

This boundary matches the exact spherical threshold of Lucibello and Mézard (2024) at every β .

Geometric picture. Fig. 1 summarises the different approaches at $\beta = 1$. The red straight line is the cusp position $\varphi_c(\alpha)$; the saddle-point capacity (19) is the load at which it reaches $\varphi = 1$. The blue solid curve is the exact-density extreme (7) (no finite capacity); the blue dotted curve is its Gaussian approximation $\sqrt{2\alpha}$, which gives the β -independent $\alpha_c^G = 1/2$. The grey dashed line $1 - \alpha/\beta$ is the Ramsauer et al. condition (11); its intersection with the blue solid curve gives the Ramsauer et al. capacity, finite and β -dependent but below the saddle-point capacity at every β .

The red line is tangent to the blue solid curve at every β ; as β varies, the tangency point sweeps the entire blue solid curve. This follows from the Legendre conjugacy between $g_{\max}(\beta)$ and $\varphi_{\max}(\alpha)$, derived in Appendix B. The geometric consequence is that the cusp $\varphi_c(\alpha, \beta)$ lies above the maximum alignment $\varphi_{\max}(\alpha)$ at every α : no individual spurious pattern enters the retrieval basin $\varphi > \varphi_c$. Retrieval loss is therefore *collective*, driven by the bulk saddle of the spurious sum and not by any single spurious pattern reaching the target. In the terminology of Lucibello and Mézard (2024), the retrieval boundary lies in the *uncondensed* phase of the auxiliary REM, where exponentially many patterns dominate the noise sum, as opposed to the *condensed* phase where a single pattern dominates.

Extension to finite temperature. At $T > 0$ the entropy density (Petrova, Polyachenko, and State 2026)

$$s(\varphi) = \frac{1}{2} \ln(1 - \varphi^2), \quad (20)$$

adds to the internal energy, giving the free energy density $f(\varphi) = u(\varphi) - T s(\varphi)$, plotted in Fig. 2 for $\alpha = 0.5$, $\beta = 1$

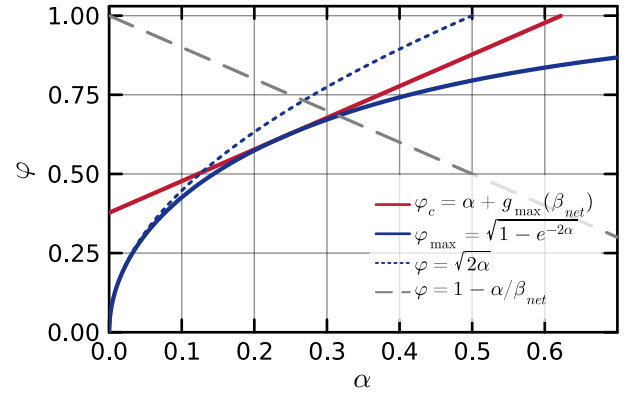


Figure 1: Threshold curves at $\beta = 1$. Red straight line: $\varphi_c(\alpha) = \alpha + g_{\max}(\beta)$, tangent to the blue solid curve at the saddle $\varphi^*(\beta)$ (Appendix B). Blue solid: the exact-density extreme alignment (7). Blue dotted: $\sqrt{2\alpha}$, its Gaussian approximation. Grey dashed: $1 - \alpha/\beta$, the Ramsauer et al. condition.

and several T . The profile has two minima: one at $\varphi \approx 0$ from the spurious sum, one at $\varphi \approx 1$ from the retrieval term. The retrieval minimum shifts from $\varphi = 1$ at $T = 0$ to

$$\varphi_{\text{eq}}(T) = \frac{1}{2}(\sqrt{T^2 + 4} - T) \quad (21)$$

at $T > 0$, and disappears at the spinodal $\varphi_{\text{eq}}(T) = \varphi_c(\alpha, \beta)$, giving

$$T_{\text{sp}}(\alpha, \beta) = \frac{\beta^2 - (\alpha + g_{\max}(\beta))^2}{\beta(\alpha + g_{\max}(\beta))}. \quad (22)$$

Inverting (22) at fixed T gives the spinodal capacity

$$\alpha_c(\beta, T) = \beta \varphi_{\text{eq}}(T) - g_{\max}(\beta). \quad (23)$$

The saddle-dominated capacity is the load at which the retrieval free energy $f_{\text{ret}}(T) = (1 - \varphi_{\text{eq}}) - \frac{T}{2} \ln(1 - \varphi_{\text{eq}}^2)$ equals the spurious-basin value, giving

$$\alpha_c^{\text{sd}}(\beta, T) = \beta[1 - f_{\text{ret}}(T)] - g_{\max}(\beta). \quad (24)$$

Both (23) and (24) reduce to (19) at $T = 0$.

Stability and Kramers escape. For $T < T_{\text{sp}}(\alpha, \beta)$ the retrieval minimum exists. It is thermodynamically stable when it is the global minimum of f and metastable when the spurious-basin minimum is deeper. At equilibrium the deeper basin carries the larger weight, with the ratio between the two basins set by their depth difference; at finite T and finite N both basins can carry significant mass. The chain escapes the retrieval basin to the spurious basin by a Kramers process at the same rate $\sim e^{-N\Delta f/T}$ in both regimes (Kramers 1940; Hänggi, Talkner, and Borkovec 1990), with Δf the per-spin barrier from the basin minimum to the cusp; what differs is the return rate. A Monte Carlo ensemble initialised in the basin contains both populations at finite budget, in proportions set by the budget over the Kramers escape time. As $T \rightarrow 0$ the escape rate vanishes, so retrieval is retained whenever the basin exists.

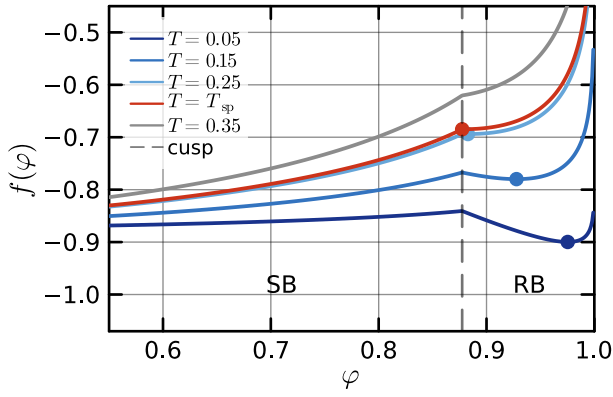


Figure 2: Leading-order free energy density $f(\phi)$ at $\alpha = 0.5$, $\beta = 1$, for several T . RB: retrieval basin; SB: spurious basin (where the spurious term dominates); cusp at $\phi \approx 0.877$ (dashed grey). Coloured dots mark the RB minima; they disappear at the spinodal temperature $T_{\text{sp}}(\alpha, \beta) \approx 0.262$.

3 Monte Carlo schemes

The main issue of Monte Carlo simulations for validation of retrieval zones of the Dense Associative Memory with exponential capacity is the computational cost of evaluating the energy (1) at each step, which involves a sum over $M = e^{\alpha N}$ patterns. Thus, in addition to the standard Metropolis scheme that evaluates the full sum (e.g., Petrova 2026), we introduce two approximations that reduce the cost per step, and we compare all three against the theoretical predictions. Below we assume $\beta = 1$.

Full. All $M = \lceil e^{\alpha N} \rceil$ patterns are generated explicitly per disorder realization and the log-sum is evaluated over all of them at every step. Cost per step is $\mathcal{O}(MN)$, dominated by the dense $M \times N$ matrix-vector product that yields the M pattern-state alignments; we batch chains into a single matrix-matrix product to saturate the GPU. M is fixed at the maximum the GPU memory allows, and N is then obtained from $N(\alpha) = \lceil \log M / \alpha \rceil$. At each (α, T) we draw 32 disorder realizations and follow two Metropolis replicas per realization, both initialized at the target ξ^1 . Step size is $\sigma = 2.4 T / \sqrt{N}$. Each replica is equilibrated for $N_{\text{eq}} = 2^{15}$ steps and sampled for $N_{\text{samp}} = 2^{13}$ steps. Per cell we record the time-averaged target alignments ϕ_a, ϕ_b and the maximum non-target alignment.

Cone. This approximation is based on our finding that the spurious basin is formed by the patterns near the saddle point. Thus, we introduce an alignment cutoff ϕ_{keep} that partitions the pattern set into two pieces. Patterns with $\phi_{1\mu} \geq \phi_{\text{keep}}$ lie inside a spherical cone of half-angle $\arccos(\phi_{\text{keep}})$ around the target ξ^1 ; these are generated explicitly and enter the log-sum term-by-term. Patterns with $\phi_{1\mu} < \phi_{\text{keep}}$ make up the passive bulk and are absorbed into a single x -independent constant

$$\ln C_{\text{bulk}} = \alpha N + \ln \int_{-1}^{\phi_{\text{keep}}} h(\phi) e^{\beta N \phi} d\phi, \quad (25)$$

which sits inside the log-sum on equal footing with the in-cone patterns. The number of in-cone patterns per disorder is a Poisson draw with mean $M p_{>}$, where $p_{>} = \Pr(\phi \geq \phi_{\text{keep}})$ under the spherical density (4). In the rescaled variable $y = (1 + \phi)/2$, the spherical density becomes a symmetric Beta and each in-cone pattern's alignment with ξ^1 is drawn from the truncated distribution

$$y \sim \text{Beta}\left(\frac{N-1}{2}, \frac{N-1}{2}\right), \quad (26)$$

together with a uniform perpendicular direction on the $(N-2)$ -sphere orthogonal to ξ^1 . The cone set is fixed once per disorder. For our ϕ_{keep} (see Results), the explicit pattern set handled per step is smaller than M by orders of magnitude; the rest enters through the closed-form bulk integral. Disorder count, replicas, initial condition, step size, equilibration and sampling lengths, and recorded observables are identical to Full. We have verified Cone-Full equivalence at a shared budget (see Results).

Bulk. No patterns are generated. The chain evolves under the leading-order analytical free energy of Section 2, with retrieval basin and spurious basin both produced by one closed-form profile $f(\phi)$. Here $\phi^1 = \xi^1 \cdot x/N$ (2) is the chain's alignment with the target. The spurious sum is replaced by its disorder-averaged saddle value (16), $e^{N(\alpha + g_{\text{max}}(\beta) - \beta)}$, an x -independent constant, and the soft-max sum (8) reduces to two terms,

$$\ln S(\phi^1) = \ln[e^{\beta N(\phi^1 - 1)} + e^{N(\alpha + g_{\text{max}}(\beta) - \beta)}], \quad (27)$$

at $\mathcal{O}(N)$ cost per step.

The Full and Cone rule $\sigma = 2.4 T / \sqrt{N}$ shrinks as $1/\sqrt{N}$, giving microscopic steps at Bulk's large- N reach: at $N = 10^4$ the chain cannot traverse the basin in $T_{\text{MC}} = 10^5$ steps, let alone resolve the Kramers escape rate that sets the boundary. We use the curvature-matched step

$$\gamma(\phi) = f''(\phi) = \frac{T(1 + \phi^2)}{(1 - \phi^2)^2}, \quad \sigma^2 = \frac{2T}{\gamma(\phi_{\text{eq}})} = \frac{2(1 - \phi_{\text{eq}}^2)^2}{1 + \phi_{\text{eq}}^2}, \quad (28)$$

where $\gamma(\phi)$ is the second derivative of the leading-order free energy density f and $\gamma_{\text{eq}} \equiv \gamma(\phi_{\text{eq}})$ is its value at the basin minimum. The explicit T in σ cancels against T in γ ; the only remaining dependence is through $\phi_{\text{eq}}(T)$, and σ is N -independent. $\gamma(\phi)$ diverges at the sphere edge $\phi = \pm 1$, so the curvature-matched step $\sqrt{2T/\gamma(\phi)}$ vanishes there: a chain at $\phi^1 = 1$ has no admissible step under this rule. We therefore initialize at the basin minimum, $\phi^1 = \phi_{\text{eq}}(T)$ (21), with perpendicular components uniform on the sphere of radius $\sqrt{N} \sqrt{1 - \phi_{\text{eq}}^2}$; there γ is finite, σ is curvature-matched by construction, and acceptance is ≈ 0.37 at every N .

We run Bulk at $N \in \{100, 300, 10^4\}$. Per α we scan T downward from $\min(T_{\text{sp}}(\alpha), 0.495)$ in steps of 0.01, running 64 independent chains of $T_{\text{MC}} = 10^5$ steps each, and record $T_{\text{bd}}(\alpha)$ as the largest T at which the residual $\phi_{\text{eq}}(T) - \langle \phi^1 \rangle$ first stays below 0.02 for three consecutive T -steps. There is no disorder.

What each scheme tests. The three schemes validate different things. Full uses all M patterns explicitly and serves as the ground-truth reference; the only approximation is Monte Carlo noise. Cone drops the patterns with $\varphi_{1\mu} < \phi_{\text{keep}}$ (almost all of M , with near-zero alignment to the target) and replaces them by a single constant from the analytical bulk integral (25); Cone matching Full at a shared M validates this analytical bulk integral to within MC noise. Bulk drops all patterns and uses the leading-order value $g_{\text{max}}(\beta)$ in $\ln S$ (27). It runs at $\mathcal{O}(N)$ cost per step and reaches N much larger than Full or Cone can; the convergence of its boundary to the spinodal as N grows validates the leading-order Kramers picture in the asymptotic limit.

4 Results

A single NVIDIA RTX A6000 GPU card with 50 Gb memory allows $M = 4.4 \times 10^6$ patterns for the Full scheme (Float32), resulting in $N(\alpha) = \lceil \log M/\alpha \rceil$ from $N = 76$ at $\alpha = 0.20$ to $N = 24$ at $\alpha = 0.65$, with a full (α, T) sweep taking about 26 days. Cone scheme at $M = 3.0 \times 10^8$ with $\phi_{\text{keep}} = 0.40$ keeps the typical in-cone count K below 1.3% of M at every cell, resulting in N from 98 to 30 over the same α range, and sweep taking in about 7 days. Bulk has no patterns and runs at $\mathcal{O}(N)$ per step on a 16-thread CPU, reaching $N \in \{100, 300, 10^4\}$ in about one, five, and ninety minutes per sweep. We have verified Cone-Full equivalence at the shared budget $M = 4.4 \times 10^6$.

Fig. 3 shows the (α, T) stability diagram at $\beta = 1$, from the Cone Monte Carlo sampler at two budgets, $M = 4.4 \times 10^6$ and $M = 3.0 \times 10^8$. The heatmap encodes the residual $\varphi_{\text{eq}}(T) - \langle \varphi \rangle$ from the $M = 3.0 \times 10^8$ run: white where the chain remains in the retrieval basin, red where it escapes to the spurious basin. The blue staircases trace, for each α column, the lowest T at which roughly 5% of chain endpoints have escaped from the retrieval basin to the spurious basin. Operationally, this is the threshold $\varphi_{\text{eq}}(T) - \langle \varphi \rangle > 0.02$ at the basin-to-saddle gap $\Delta\varphi \approx 0.40$ along the boundary, after a light smoothing pass to suppress single-cell MC noise. At fixed α , increasing the budget from $M = 4.4 \times 10^6$ to $M = 3.0 \times 10^8$ (which raises the reachable N via the N -ramp) shifts the boundary to higher T . The expected destination at $N \rightarrow \infty$ is the spinodal $T_{\text{sp}}(\alpha, \beta)$ (22): below the spinodal the retrieval basin still exists, and a chain initialised in the basin escapes only on the Kramers time $\tau_K \sim e^{N\Delta f/T}$, which grows exponentially with N . At the N values reached by the Cone sampler here ($N \leq 98$) the boundary is still far from the spinodal. It happens to lie close to the saddle line $\alpha_c^{\text{sd}}(\beta, T)$ (24), but this proximity is incidental: as established in Section 2, α_c^{sd} is a thermodynamic crossover, not a sharp boundary, with no sharp change in the retrieval fraction as we cross it. The Gaussian prediction $\alpha_c^{\text{G}}(T)$ sits to the left of both Cone boundaries, underestimating the capacity at every T .

Finite- N residual basin above the LO boundary. At low T the Cone boundary in Fig. 3 sits to the right of the zero-temperature capacity $\alpha_c(\beta, 0) = \beta - g_{\text{max}}(\beta) = 0.6226$ at $\beta = 1$, reaching $\alpha \approx 0.70$. This is a finite- N effect. The softmax sum (8), with the spurious term replaced by its Laplace

value (40) from Appendix A, defines the finite- N effective potential at fixed disorder,

$$\frac{V_N(\phi)}{N} = 1 - \frac{1}{\beta N} \ln \left[e^{\beta N \phi} + \frac{e^{\beta N \varphi_c(\alpha, \beta)}}{(1 - \varphi^{*4}(\beta))^{1/2}} \right] - \frac{T(N-3)}{2N} \ln(1 - \phi^2), \quad (29)$$

where the entropy coefficient $(N-3)/(2N)$ comes from the exact spherical marginal $h(\phi) \propto (1 - \phi^2)^{(N-3)/2}$ in (4), not its $N \rightarrow \infty$ limit $1/2$. The $N \rightarrow \infty$ limit of (29) is the LO landscape

$$\frac{V_N(\phi)}{N} \rightarrow 1 - \max(\phi, \varphi_c) - \frac{T}{2} \ln(1 - \phi^2), \quad (30)$$

and at finite N the log-sum smooths the LO corner over width $(\beta N)^{-1}$. Whenever the cusp φ_c sits just above 1 (the regime $\alpha > \alpha_c(\beta, 0)$), the smoothing reaches onto the sphere edge and produces a local minimum of V_N at ϕ just below 1 that the LO curve does not have. Fig. 4 shows this at the three loads $\alpha = 0.62, 0.65, 0.70$ for the $N(\alpha)$ at $M = 4.4 \times 10^6$.

The local-minimum position of V_N matches the steady-state alignment of cold-initialised chains from the Full/Cone run in the same (α, T) cells to within 0.002. For $\alpha = 0.62, 0.65$ it is the global minimum of V_N ; for $\alpha = 0.70$ it is metastable with a small per-chain Boltzmann exponent (≈ 1.3 at $T = 0.005$) and no chain in the Full/Cone ensembles escapes within T_{MC} . Setting $V'_N = V''_N = 0$ gives a finite- N correction to the zero-temperature capacity (Appendix C),

$$\alpha_c^{\text{fN}}(\beta, T, N) = \alpha_c(\beta, 0) - \frac{1 + \ln(T\beta(N-3)/[2\sqrt{1-\varphi^{*4}}])}{N}, \quad (31)$$

positive for $T\beta(N-3) < 2\sqrt{1-\varphi^{*4}}/e$ and vanishing as $1/N$ at fixed T . At $\beta = 1, T = 0.005$ this gives $\alpha_c^{\text{fN}} = 0.712$ at $N(0.70) = 22$ (Cone $M = 4.4 \times 10^6$) and 0.683 at $N(0.70) = 28$ (Cone $M = 3 \times 10^8$): the right edge of the heatmap retracts from $\alpha \approx 0.71$ at the small budget to $\alpha \approx 0.68$ at the large budget, matching the empirical 5% blue staircases in Fig. 3.

The green curves trace the Bulk boundary at the same threshold, $T_{\text{MC}} = 10^5$ steps per chain, and $N \in \{100, 300, 10^4\}$. Bulk carries no live patterns ($K = 0$) and the chain follows the two-term log-sum (27) directly, so the line is the pure Kramers-escape boundary from the basin minimum at finite N . Empirically the gap to the spinodal shrinks as $1/\sqrt{N}$: at $\alpha = 0.50$ the gap drops from $\Delta T \approx 0.13$ at $N = 100$ to 0.08 at $N = 300$, and by $N = 10^4$ the boundary sits within one T -grid cell of the spinodal across $\alpha \in [0.40, 0.62]$ and is visually indistinguishable from it (the spinodal in the figure is drawn last, on top of the lightest green $N = 10^4$ curve, by design). The march of the Bulk boundary onto the spinodal as N grows is the asymptotic claim the theory makes, now visible across two decades of N in a single figure.

Dependence of the Bulk boundary on the MC budget is much weaker than on N . Solving the 5% criterion $T_{\text{MC}} k =$

0.05 for the Kramers rate $k = (1/2\pi)e^{-N\Delta f/T}$ gives $N\Delta f/T = \ln T_{\text{MC}} + O(1)$, so the gap to the spinodal grows only as $\sqrt{\ln T_{\text{MC}}}$: about 9% per decade in T_{MC} from the present $T_{\text{MC}} = 10^5$, against the $\sqrt{10} \approx 3.16$ shrinkage per decade in N . Concretely, going from $T_{\text{MC}} = 10^5$ to 10^6 would shift the $N = 100$ Bulk line by $\Delta T_{\text{bd}} \approx 0.01$ – 0.04 at $T \geq 0.2$ and by less than the T -grid spacing $dT = 0.01$ at $T \leq 0.05$. Near the LSE capacity tip $\alpha_{\text{LSE}} \approx 0.6226$ the spinodal itself collapses to $T_{\text{sp}} = 0$, so the absolute gap budget is small and T_{MC} shifts there are pinned below the T -grid spacing. Convergence to the spinodal in the figure is therefore controlled by N ; multiplying the per-chain budget by orders of magnitude would barely move any of the three green curves.

5 Discussion

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A Appendix A: Derivation of the prefactor in the spurious sum

The integral (12) evaluates in closed form. Substituting $\varphi = \cos \theta$ maps the spherical density onto the kernel of the integral representation of the modified Bessel function $I_\nu(z)$ (Gradshteyn and Ryzhik 2015, 8.431.3); at $\nu = N/2 - 1$, $z = \beta N$, this gives for $N > 1$

$$S_{\text{spur}} = M\Gamma\left(\frac{N}{2}\right) \left(\frac{2}{\beta N}\right)^{N/2-1} e^{-\beta N} I_{N/2-1}(\beta N). \quad (32)$$

The right-hand side is the disorder expectation of the sum (9). The relative fluctuations of S_{spur} around this expectation are $\sim e^{N[\delta(\beta)-\alpha]/2}$ with $\delta(\beta) = g_{\text{max}}(2\beta) - 2g_{\text{max}}(\beta) > 0$; they are exponentially small in N when $\alpha > \delta(\beta)$. At $\beta = 1$, $\delta \approx 0.334$, well below the retrieval boundary $\alpha_c \approx 0.623$, so the saddle (16) captures the typical S_{spur} .

The Debye uniform asymptotic of the modified Bessel function (Olver et al. 2010, 10.41.3) gives the prefactor of (16). Keeping the leading ($k = 0$) term of the asymptotic series,

$$I_\nu(\nu z') \sim \frac{e^{\nu\eta(z')}}{(2\pi\nu)^{1/2}(1+z'^2)^{1/4}}, \quad (33)$$

where $z' = 2\beta N/(N-2)$ and

$$\eta(z') = \sqrt{1+z'^2} + \ln \frac{z'}{1+\sqrt{1+z'^2}}. \quad (34)$$

$\eta(z')$ is expanded to $\mathcal{O}(N^{-1})$ because it is multiplied by $\nu = N/2 - 1$ in the exponent:

$$\eta(z') = \eta(2\beta) + \eta'(2\beta) \frac{4\beta}{N} + \mathcal{O}(N^{-2}), \quad (35)$$

where

$$\eta(2\beta) = 1 + 2\beta\varphi^* + \ln \varphi^*, \quad (36)$$

$$\eta'(2\beta) = \varphi^* + \frac{1}{2\beta}. \quad (37)$$

Then

$$\nu\eta(z') = \frac{N}{2}\eta(2\beta) - \ln \varphi^* + \mathcal{O}(1/N). \quad (38)$$

The $\mathcal{O}(1)$ piece $-\ln \varphi^*$ in the exponent contributes a factor $1/\varphi^*$ to the prefactor. Using Stirling's formula

$$\Gamma(N/2) \sim \sqrt{4\pi/N} (N/2)^{N/2} e^{-N/2} \quad (39)$$

and collecting all terms in (32) to $\mathcal{O}(1)$ in the exponent and to leading order in the prefactor reproduces the exponent of (16) and gives the prefactor:

$$S_{\text{spur}} \approx \frac{e^{N[\alpha+g_{\text{max}}(\beta)-\beta]}}{\sqrt{1-\varphi^{*4}}} \quad (40)$$

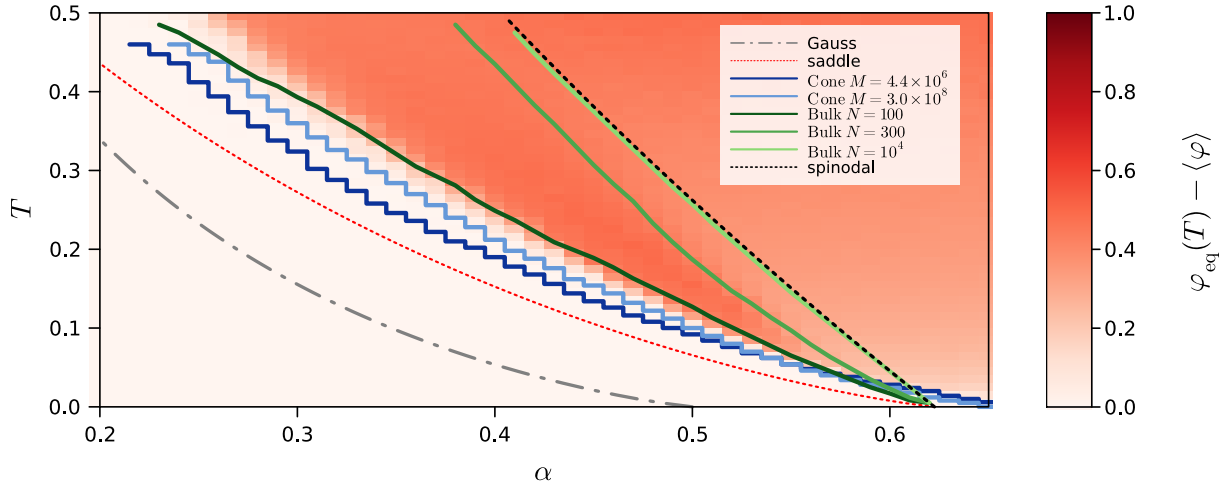


Figure 3: (α, T) stability diagram at $\beta = 1$. Heatmap: $\varphi_{\text{eq}}(T) - \langle \varphi \rangle$ from the $M = 3.0 \times 10^8$ Cone MC run. Blue staircases: Cone retrieval boundary at threshold 0.02 (roughly 5% of chain endpoints escaped) for $M = 4.4 \times 10^6$ (dark) and $M = 3.0 \times 10^8$ (light). Green curves: Bulk retrieval boundary at $T_{\text{MC}} = 10^5$ and $N \in \{100, 300, 10^4\}$, dark to light with N . The light-green $N = 10^4$ curve sits on the spinodal across most of the α range; the spinodal is drawn last to make this coincidence visible. Reference curves: saddle boundary $\alpha_c^{\text{sd}}(\beta, T)$ (24) (red dotted), spinodal T_{sp} (22) (black dotted), Gaussian baseline $\alpha_c^{\text{G}}(T)$ (grey dash-dotted).

with $g_{\text{max}}(\beta)$ given by (15). At $\beta = 1$, $\varphi^* \approx 0.618$ and the prefactor is ≈ 1.082 .

B Appendix B: Tangency of the cusp lines to the spherical extreme

To show that the cusp lines (18) are tangent to the spherical-extreme curve (7) at every β , we need to show that the slope and the φ -intercept coincide.

Differentiating (7) once gives

$$\varphi'_{\text{max}}(\alpha) = \frac{1 - \varphi_{\text{max}}^2}{\varphi_{\text{max}}}, \quad (41)$$

and a second differentiation gives $\varphi''_{\text{max}}(\alpha) < 0$ for $\alpha > 0$, so φ_{max} is strictly concave on $(0, \infty)$. For a concave function every tangent line lies above the function, with equality only at the tangency point. This is used in the lower-envelope identity below.

The cusp line at sharpness β ,

$$\varphi = \alpha/\beta + g_{\text{max}}(\beta)/\beta, \quad (42)$$

has slope $1/\beta$ and φ -intercept $g_{\text{max}}(\beta)/\beta$. To find the point on the blue curve where the tangent has the same slope, set the right-hand side of (41) equal to $1/\beta$:

$$\frac{1 - \varphi^2}{\varphi} = \frac{1}{\beta} \iff \beta \varphi^2 + \varphi - \beta = 0. \quad (43)$$

This is the saddle equation and its unique positive root is the spherical saddle (14). The corresponding α on the blue curve follows from $\varphi^* = \varphi_{\text{max}}(\alpha^*)$:

$$\alpha^*(\beta) = -\frac{1}{2} \ln(1 - \varphi^{*2}). \quad (44)$$

A line of slope $1/\beta$ passing through $(\alpha^*(\beta), \varphi^*(\beta))$ has φ -intercept

$$b_{\text{tg}}(\beta) = \varphi^* - \alpha^*/\beta. \quad (45)$$

The cusp line has φ -intercept $g_{\text{max}}(\beta)/\beta$. Using (15), the φ -intercepts coincide; the two lines are therefore identical.

By the concavity statement above, the cusp line at any sharpness β lies above the blue curve, with equality only at $\alpha = \alpha^*(\beta)$:

$$\varphi_c(\alpha, \beta) \geq \varphi_{\text{max}}(\alpha) \quad \text{for all } \alpha \text{ and all } \beta > 0, \quad (46)$$

with equality at one α for each β . Taking the infimum over β ,

$$\varphi_{\text{max}}(\alpha) = \inf_{\beta > 0} \varphi_c(\alpha, \beta), \quad (47)$$

the lower-envelope identity.

The saddle function $\varphi^*(\beta)$ (14) is continuous and strictly increasing on $(0, \infty)$ from 0 to 1. As β sweeps $(0, \infty)$, the tangency point $(\alpha^*(\beta), \varphi^*(\beta))$ therefore traces the entire blue curve from the origin to its right edge.

Two results follow from the geometric setup above. At $T = 0$ the cusp line of slope $1/\beta$ reaches $\varphi = 1$ at $\alpha = \beta - g_{\text{max}}(\beta)$, which is the zero-temperature capacity (19) (equivalently, the spinodal at $T = 0$).

Ramsauer's bound is too restrictive. The Ramsauer et al.'s condition $\varphi_{\text{max}}(\alpha) = 1 - \alpha/\beta$ is the intersection of φ_{max} with the line of slope $-1/\beta$ through $(0, 1)$. By (47), φ_{max} lies below every cusp line, so the Ramsauer et al.'s line crosses φ_{max} at strictly smaller α than the cusp line at slope $1/\beta$ reaches $\varphi = 1$. Ramsauer et al.'s capacity α_c^R therefore satisfies $\alpha_c^R < \alpha_c(\beta, 0)$ for every $\beta > 0$.

Asymptotic capacity. Using (14) in (15) and then in (19), one obtains:

$$\begin{aligned} \alpha_c(\beta, 0) &= \beta - g_{\text{max}}(\beta) \rightarrow \beta & (\beta \rightarrow 0), \\ \alpha_c(\beta, 0) &= \beta - g_{\text{max}}(\beta) \rightarrow \frac{1}{2} \ln \beta + \frac{1}{2} & (\beta \rightarrow \infty), \end{aligned} \quad (48)$$

the small- and large- β limits of the zero-temperature capacity cited in the Abstract.

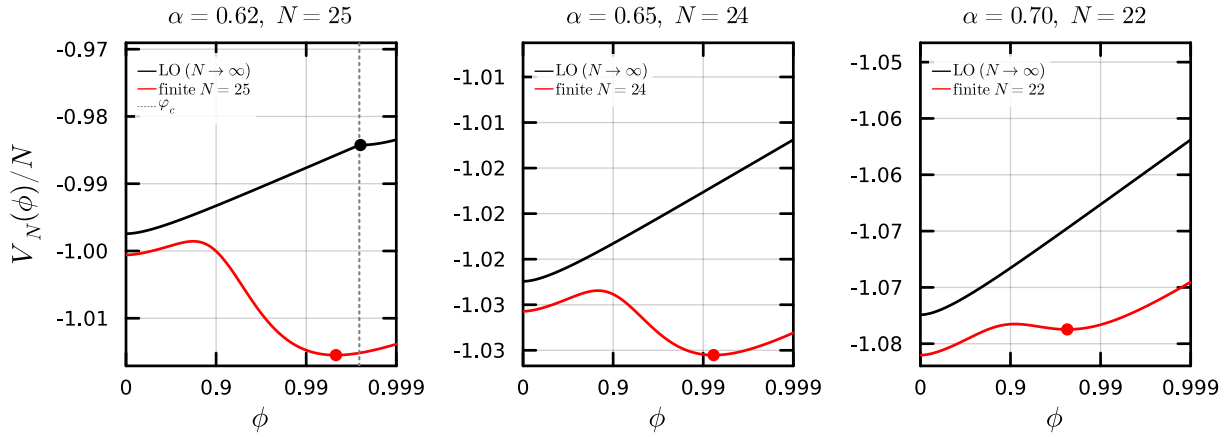


Figure 4: Finite- N effective potential $V_N(\phi)/N$ at $\beta = 1$, $T = 0.005$, against ϕ on a $\lg(1 - \phi)$ axis for three loads above $\alpha_c(\beta, 0) = 0.6226$. Black: leading-order landscape $-\max(\phi, \varphi_c) - (T/2) \log(1 - \phi^2)$. Red: the finite- N form (29) at $N(\alpha)$ for the Cone run at $M = 4.4 \times 10^6$. The log-sum smooths the LO corner over width $\sim 1/(\beta N)$, generating a local minimum near $\phi = 1$ that the LO curve lacks. Red dots: V_N residual-basin minima. Black dot (left panel only): LO retrieval-basin minimum at $\phi = \varphi_{\text{eq}}(T)$, present only when $\varphi_c < 1$. Grey dotted: LO cusp position φ_c , shown only when $\varphi_c < 1$.

Legendre transform. The lower-envelope identity (47) and the formula below are the two sides of a Legendre transform between $\varphi_{\text{max}}(\alpha)$ and $g_{\text{max}}(\beta)/\beta$. The forward transform takes φ_{max} to its conjugate,

$$g_{\text{max}}(\beta)/\beta = \sup_{\alpha} [\varphi_{\text{max}}(\alpha) - \alpha/\beta], \quad (49)$$

with the supremum attained at $\alpha = \alpha^*(\beta)$ from the slope-match condition (43). The Legendre transform is involutive: applying it to $g_{\text{max}}(\beta)/\beta$ recovers $\varphi_{\text{max}}(\alpha)$, which is exactly (47) after using the cusp-line definition (18).

Two facts follow from the duality. First, the lower-envelope identity (47) is the inverse of (49), so the tangency is automatic. Second, differentiating (15) in β and using the envelope theorem $\partial g / \partial \varphi|_{\varphi^*} = 0$ gives

$$g'_{\text{max}}(\beta) = \varphi^*(\beta), \quad (50)$$

so $g_{\text{max}}(\beta)$ is recovered by integrating $\varphi^*(\beta)$. Plugging the small- and large- β limits $\varphi^* \sim \beta$ and $\varphi^* \sim 1 - 1/(2\beta)$ into (50) and integrating reproduces (48).

C Appendix C: Finite- N correction to the zero-temperature capacity

We derive the finite- N correction (31) quoted in the Results. Differentiating (29) in ϕ gives

$$\frac{1}{N} \frac{dV_N}{d\phi} = -\varsigma\left(\beta N(\phi - \varphi_c) + \frac{1}{2} \ln(1 - \varphi^{*4})\right) + \frac{T(N-3)\phi}{N(1 - \phi^2)}, \quad (51)$$

with $\varsigma(x) = 1/(1 + e^{-x})$ the logistic and $\varphi_c = (\alpha + g_{\text{max}}(\beta))/\beta$ from (18). The $\frac{1}{2} \ln(1 - \varphi^{*4})$ shift inside the logistic is the $\ln$ of the Appendix A prefactor. The factor $(N-3)/N$ on the entropy slope comes from the exact spherical marginal (4), kept unexpanded. A critical point of V_N balances the smoothed jump against the sphere-entropy slope.

For $\varphi_c > 1$ (equivalently $\alpha > \alpha_c(\beta, 0)$) the LO floor $-\varphi_c$ sits off the unit sphere; the smoothing still reaches a neighbourhood of $\phi = 1$ where the entropy slope diverges, so the balance holds in a window of width $\sim 1/(\beta N)$ below $\phi = 1$ and admits two solutions (the local minimum and the adjacent barrier saddle) that merge as N decreases. At the merger, $V'_N = V''_N = 0$ simultaneously; in the regime $T\beta(N-3) \ll 1$ this two-equation system reduces to

$$\beta N(\varphi_c - 1) + \ln\left(\frac{T\beta(N-3)}{2\sqrt{1 - \varphi^{*4}}}\right) + 1 = 0. \quad (52)$$

Solving (52) for φ_c and substituting $\varphi_c = (\alpha + g_{\text{max}})/\beta$ together with $\alpha_c(\beta, 0) = \beta - g_{\text{max}}(\beta)$ (19) gives (31). For $\varphi_c \leq 1$ no bifurcation occurs: the LO basin already sits on the unit sphere and the residual minimum exists for all N .

Numerical values at $\beta = 1$, $T = 0.005$ ($\varphi^* \approx 0.618$, prefactor $1/\sqrt{1 - \varphi^{*4}} \approx 1.082$): $\alpha_c^{\text{fn}}(22) = 0.712$, $\alpha_c^{\text{fn}}(24) = 0.700$, $\alpha_c^{\text{fn}}(28) = 0.683$, $\alpha_c^{\text{fn}}(30) = 0.676$, $\alpha_c^{\text{fn}}(100) = 0.626$. The Cone run at $M = 4.4 \times 10^6$ reaches $N(0.65) = 24$ and $N(0.70) = 22$; both are below the local thresholds, so the residual basin is visible at both loads in Fig. 4. The Cone run at $M = 3 \times 10^8$ reaches $N(0.65) = 30$ and $N(0.70) = 28$: at $\alpha = 0.65$ the basin survives ($\alpha_c^{\text{fn}}(30) = 0.676 > 0.65$), at $\alpha = 0.70$ it disappears ($\alpha_c^{\text{fn}}(28) = 0.683 < 0.70$).